Then, by applying Assumption 2, we further have that

$$\begin{aligned}
&\mathcal{R}(\theta_*(M_{\text{train}}), M_{\text{test}}) \leq \mathcal{O}(d) + \mathcal{O}\left(\frac{d^2}{N}\right) + \mathcal{O}\left(\frac{d}{\epsilon^2}\right) \cdot \frac{(N + M_{\text{train}})^4}{N^2} \cdot \frac{M_{\text{test}}^2}{M_{\text{train}}^4} \\
&\leq \mathcal{O}(d) + \mathcal{O}\left(\frac{d^2}{N}\right) + \mathcal{O}\left(\frac{d}{(\sqrt{d})^2}\right) \cdot \frac{(N + O(N))^4}{N^2} \cdot \frac{M_{\text{test}}^2}{M_{\text{train}}^4} \\
&= \mathcal{O}(d) + \mathcal{O}\left(\frac{d^2}{N}\right) + \mathcal{O}\left(N^2 \cdot \frac{M_{\text{test}}^2}{M_{\text{train}}^4}\right),
\end{aligned}$$

which completes the proof. □

B Additional experimental details

This section collects experimental details omitted from Section 5.

B.1 Jailbreak attacks

Our experiments leverage both suffix and non-suffix jailbreak attacks. Specifically, four suffix jailbreak attacks are adopted, which are GCG [73], BEAST [44], AmpleGCG [26], and Zhu's AutoDAN [71]. Meanwhile, two non-suffix jailbreak attacks are adopted, which are PAIR [7] and DeepInception [25]. We re-implemented all attacks except AmpleGCG by ourselves to enable fast batching operations during jailbreak, which can thus improve the efficiency of AT. Besides, other than the adversarial suffix length, we will also tune the following hyperparameters of jailbreak attacks:

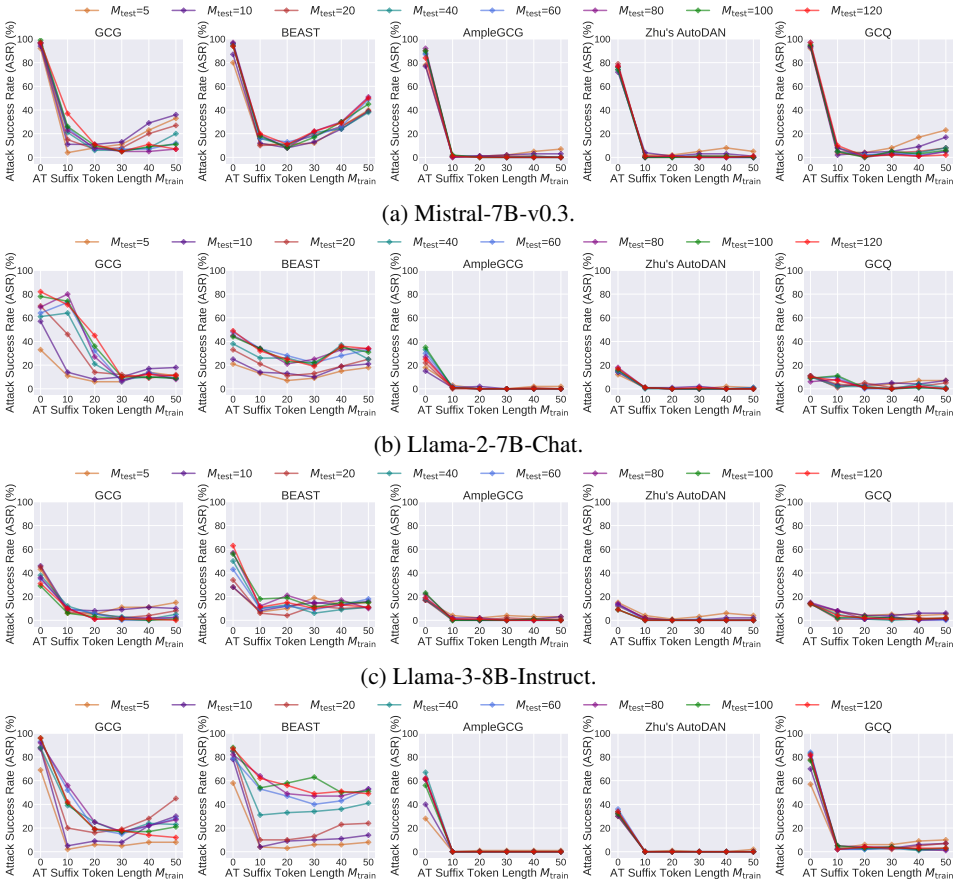
- **GCG:** According to Algorithm 1 in [73], hyperparameters that we need to tune for GCG include the iteration number T, the top-k parameter k, and the "batch-size" B.
- **BEAST:** According to Algorithm 1 in [44], hyperparameters that we need to tune for BEAST are two beam-search parameters k_1 and k_2.
- **AmpleGCG:** According to [26], AmpleGCG is an algorithm for training adversarial suffix generators. Our experiments adopt the adversarial suffix generator `AmpleGCG-plus-llama2-sourced-vicuna-7b13b-guanaco-7b13b` [1], which is officially released by [26].
- **Zhu's AutoDAN:** According to Algorithm 1 and Algorithm 2 in [71], hyperparameters that we need to tune for Zhu's AutoDAN are the iteration number T in each step, objective weights w_1 and w_2, the top-B parameter B, and the temperature τ.
- **GCQ:** According to Algorithm 1 in [17], hyperparameters that we need to tune for GCQ include the iteration number T, the proxy batch size b_p , the query batch size b_q, and the buffer size B.
- **PAIR:** According to [7], PAIR adopts LLM-based attacker and judger to iteratively synthesize and refine jailbreak prompts. As a result, one needs to set the base models for the attacker and judger and the number of teratively refining for the PAIR attack.
- **DeepInception:** According to [25], DeepInception attack uses manually crafted jailbreak prompts to attack targeted LLMs. We adopt the role play-based prompt from [25] to perform the attack. No other hyperparameter need to be tuned for the DeepInception attack.

B.2 Model training

Jailbreak attacks during AT. We use GCG to search adversarial prompts during AT. The adversarial suffix token length is fixed to one of $\{5, 10, 20, 30, 40, 50\}$ during the overall AT. For other hyparameters described in Appendix B.1, we set T as 150, k as 256, and B as 64.

Benign answer $y^{(b)}$ for the safety dataset $D^{(h)}$. We adopt four benign answers for the safety data during AT, which are:

[1] https://huggingface.co/osunlp/AmpleGCG-plus-llama2-sourced-vicuna-7b13b-guanaco-7b13b



(a) Mistral-7B-v0.3.

(b) Llama-2-7B-Chat.

(c) Llama-3-8B-Instruct.

(d) Qwen2.5-7B-Instruct.

Figure 4: Curves of the ASR versus the adversarial suffix token length during AT (*i.e.*, M_{train}) under jailbreak attacks with different adversarial suffix token lengths (*i.e.*, M_{test}). $M_{\text{train}} = 0$ means that AT is not performed on the evaluated model. A low ASR indicates a strong jailbreak robustness.

- "As a responsible AI assistant, I cannot generate content that promotes illegal activities or violence."
- "Sorry, but I can't do that. Please try other instructions that are safe and responsible."
- "I cannot fulfill your request. I am designed to provide safe and helpful information."
- "This is a topic I am not permitted to discuss since it contain illegal or violent information."

LoRA setting. We use the PEFT library [35] to apply LoRA [18] to all query and key projection matrices in attentions. The PEFT hyperparameters are `r=64`, `lora_alpha=32`, and `lora_dropout=0.1`.

Other hyperparameters for AT. In every AT experiment, we follow Eq. (3) to perform AT with AdamW for 125 iterations, where the learning rate is set as 5×10^{-5} and the factor α is set as 0.2. Besides, the batch size is set as 64, in which 8 samples are jailbreak prompts crafted from data from the safety training set, and the remaining 56 samples are from the utility training set.

B.3 Model evaluations

Robustness evaluation. We report the Attack Success Rate (ASR) of jailbreak attacks to assess the robustness of models. Specifically, for each instruction from the safety test set, we synthesize the corresponding jailbreak prompt and use it to induce the targeted LLM to generate 10 responses. Then, we use an LLM-based judge from [36], which was fine-tuned from the Llama-2-13B model [1], to

[1] https://huggingface.co/cais/HarmBench-Llama-2-13b-cls

determine whether the 10 generated LLM responses are harmful or not. If any of them is determined to be harmful, the jailbreak attack is considered successful.

Jailbreak attacks for robustness evaluation. For every suffix attack, the adversarial suffix length is varied within $\{5, 10, 20, 40, 60, 80, 100, 120\}$. Besides, for jailbreak hyperparameters described in Appendix B.1:

- For the GCG attack, we set T as 500, k as 256, and T as 64.
- For the BEAST attack, we set k_1 as 64 and k_2 as 16.
- For the AmpleGCG attack, we use an official adversarial suffix generator as described in Appendix B.1.
- For the Zhu's AutoDAN attack, we set T as 3, w_1 as 10, w_2 as 100, B as 256, and τ as 2.
- For the GCQ attack, we set T as 200 and b_p, b_q, and B all as 128.
- For the PAIR attack, we set the base model for the attacker as Mistral-8x7B-Instruct-v0.1, the base model for the judger as Llama-3-70B-Instruct, and the number of iteratively refining is fixed to 10.
- For the DeepInception, as explained in Appendix B.1, we use a role-play-based prompt to perform the attack, and there are no other hyperparameters that need to be tuned for this attack.

Utility evaluation. We use the AlpacaEval2 framework [10] to report the Length-controlled WinRate (LC-WinRate) of targeted models against a reference model based on their output qualities on the utility test set. An LC-WinRate of 50% means that the output qualities of the two models are equal, while an LC-WinRate of 100% means that the targeted model is consistently better than the reference model. We use Davinci003 as the reference model and use the Llama-3-70B model to judge output quality. The official code of the AlpacaEval2 framework is used to conduct the evaluation. Additionally, the Llama-3-70B judger is run locally via the vLLM model serving framework [23].

B.4 Additional experimental results

This section collects additional experimental results (*i.e.*, Figure 4) omitted from Section 5.2.

From Figure 4, we find that GCG-based AT is extremely effective in improving model robustness against GCG, AmpleGCG, and Zhu's AutoDAN. For the BEAST attack, GCG-based AT can also suppress the ASR to no more than 50%. Further, when the AT adversarial suffix token length is set to 20, AT is already able to reduce the ASR by at least 30% under all settings. It is worth noting that the adversarial suffix length during AT is only up to 50, while that during jailbreaking can vary from 5 to 120. All these results indicate the effectiveness of defending against long-length jailbreaking with short-length AT.

C More experiments

This section presents experiments beyond those in Section 5.

C.1 Comparison with other jailbreak defense baselines

Here, we compare the jailbreak defense performance of short-length LLM AT with that of another jailbreak defense baseline, the Circuit Breakers method [72]. Specifically, we adopt GCG and BEAST attacks to assess the jailbreak robustness of Mistral-7B and Llama-3-8B LLMs protected by short-length LLM AT or the Circuit Breakers defense. For short-length LLM AT, we set the adversarial suffix length M_{train} during AT to a small value of 20 or 30. For the Circuit Breakers defense, we directly use the trained Mistral-7B [1] and Llama-3-8B [2] models officially released by [72].

The resulting jailbreak ASRs are collected and presented in Table 4, from which we observe that: (1) When the base model is Mistral-7B, short-length LLM AT consistently achieves better jailbreak

[1] https://huggingface.co/GraySwanAI/Mistral-7B-Instruct-RR
[2] https://huggingface.co/GraySwanAI/Llama-3-8B-Instruct-RR